

B K Birla Institute of Engineering & Technology, Pilani

45 Days Summer Training Program

Subject: Data Science with AIML

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Branch: CS

College: BKBIET Pilani

Assignment 3

Question 1

Create a class named Student with attributes: name, age, and course. Create an object and print all the details.

Program:

```
class Student:
```

```
    def __init__(self, name, age, course):
```

```
        self.name = name
```

```
        self.age = age
```

```
        self.course = course
```

```
student = Student("Praveen Poonia", 18, "Data Science with AIML")
```

```
print("Name:", student.name)
```

```
print("Age:", student.age)
```

```
print("Course:", student.course)
```

Output:

Name: Praveen Poonia

Age: 18

Course: Data Science with AIML

Question 2

Create a class named Car with attributes: brand, model, and price. Create two objects and display their details.

Program:

```
class Car:
```

```
    def __init__(self, brand, model, price):
```

```
        self.brand = brand
        self.model = model
        self.price = price

car1 = Car("Tesla", "Model 3", 4000000)
car2 = Car("Mahindra", "BE 6", 2200000)

print("Car 1")
print(car1.brand, car1.model, car1.price)

print("\nCar 2")
print(car2.brand, car2.model, car2.price)
```

Output:

```
Car 1
Tesla Model 3 4000000
```

```
Car 2
Mahindra BE 6 2200000
```

Question 3

Create a class named Employee and use a constructor (init) to initialize employee_id, name, and salary. Print the employee details.

Program:

```
class Employee:
    def __init__(self, employee_id, name, salary):
        self.employee_id = employee_id
        self.name = name
        self.salary = salary

employee = Employee("EMP101", "Praveen Poonia", 50000)

print("Employee ID:", employee.employee_id)
print("Name:", employee.name)
print("Salary:", employee.salary)
```

Output:

```
Employee ID: EMP101
Name: Praveen Poonia
Salary: 50000
```

Question 4

Create a class named Rectangle with attributes length and width. Create a method to calculate and print the area of the rectangle.

Program:

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        print("Area of Rectangle =", self.length * self.width)

rect = Rectangle(10, 5)
rect.area()
```

Output:

Area of Rectangle = 50

Question 5

Create a class named Circle with an attribute radius. Create a method that calculates and prints the area of the circle.

Program:

```
class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        area = 3.14 * self.radius * self.radius
        print("Area of Circle =", area)

circle = Circle(7)
circle.area()
```

Output:

Area of Circle = 153.86

Question 6

Create a class named BankAccount with methods deposit(amount) and withdraw(amount). Display the updated balance after each transaction.

Program:

```
class BankAccount:
```

```
    def __init__(self, account_holder, balance):
```

```
        self.account_holder = account_holder
```

```
        self.balance = balance
```

```
    def deposit(self, amount):
```

```
        self.balance += amount
```

```
        print("Balance after deposit:", self.balance)
```

```
    def withdraw(self, amount):
```

```
        self.balance -= amount
```

```
        print("Balance after withdrawal:", self.balance)
```

```
account = BankAccount("Praveen Poonia", 10000)
```

```
account.deposit(2000)
```

```
account.withdraw(1500)
```

Output:

Balance after deposit: 12000

Balance after withdrawal: 10500

Question 7

Create a class named Animal with a method sound(). Create a child class Dog that overrides the sound() method and prints "Bark".

Program:

```
class Animal:
```

```
    def sound(self):
```

```
        print("Animal Sound")
```

```
class Dog(Animal):
```

```
    def sound(self):
```

```
        print("Bark")
```

```
dog = Dog()
```

```
dog.sound()
```

Output:

Bark

Question 8

Create a class named Person with attributes name and age. Create a child class Student with an additional attribute roll_number. Display all details using a Student object.

Program:

```
class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

class Student(Person):
    def __init__(self, name, age, roll_number):
        super().__init__(name, age)
        self.roll_number = roll_number

student = Student("Praveen Poonia", 18, "24EBKCS087")

print("Name:", student.name)
print("Age:", student.age)
print("Roll Number:", student.roll_number)
```

Output:

```
Name: Praveen Poonia
Age: 18
Roll Number: 24EBKCS087
```

Question 9

Create a class named Calculator with methods add(), subtract(), multiply(), and divide(). Take two numbers as input and perform all operations.

Program:

```
class Calculator:
    def __init__(self, num1, num2):
        self.num1 = num1
        self.num2 = num2

    def add(self):
        print("Addition =", self.num1 + self.num2)

    def subtract(self):
        print("Subtraction =", self.num1 - self.num2)

    def multiply(self):
```

```
print("Multiplication =", self.num1 * self.num2)

def divide(self):
    print("Division =", self.num1 / self.num2)

calc = Calculator(20, 5)

calc.add()
calc.subtract()
calc.multiply()
calc.divide()
```

Output:

```
Addition = 25
Subtraction = 15
Multiplication = 100
Division = 4.0
```

Question 10

Create a class named `LibraryBook` with attributes `book_name`, `author`, and `price`. Create a method `display_book_info()` that prints all book details. Create at least three book objects and display their information.

Program:

```
class LibraryBook:
    def __init__(self, book_name, author, price):
        self.book_name = book_name
        self.author = author
        self.price = price

    def display_book_info(self):
        print("Book Name:", self.book_name)
        print("Author:", self.author)
        print("Price:", self.price)
        print()

book1 = LibraryBook("Rich Dad Poor Dad", "Robert Kiyosaki", 350)
book2 = LibraryBook("Atomic Habits", "James Clear", 450)
book3 = LibraryBook("The Psychology of Money", "Morgan Housel", 399)

book1.display_book_info()
book2.display_book_info()
book3.display_book_info()
```

Output:

Book Name: Rich Dad Poor Dad

Author: Robert Kiyosaki

Price: 350

Book Name: Atomic Habits

Author: James Clear

Price: 450

Book Name: The Psychology of Money

Author: Morgan Housel

Price: 399